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New insights about the secondary–tertiary transition in mathematics

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Abstract
In this concluding commentary, I embed the articles of the special issue "Transition from school into university mathematics: experiences across educational contexts" into the current landscape of research on secondary–tertiary transition, and discuss the advancement they made related to the synthesis published in 2008 by Gueudet. My aim is to elaborate a new synthesis across the topics of affect and the use of sociocultural perspectives that have framed the work of all contributions. For that, I focus on four major themes. First, I consider the complex relationship between affect and socio-cultural dimensions in the papers, which raises the need for theory networking in further research clarifying mutual roles of the two dimensions. Second, international diversity is only partially expressed in the collection of papers, thus should be a major topic of research in future that would provide a more comprehensive view on the transition from school to university mathematics. I thirdly discuss how the collection of papers deepens our understanding of the complexity and diversity related to the students' experience with the secondary–tertiary transition and its potential impact for teaching. Finally, I propose and justify the need for systematic design and evaluation of interventions targeting a change in students' affect related to the transition. In sum, the wide range of the significant findings in this special issue and their mutual relations add new perspectives to the research on secondary–tertiary transition and evidence a new spectrum of research directions.

Keywords Affect · Enculturation in mathematical practices · Secondary–tertiary transition · Socio-cultural theories

The low success rate in the first year of post-secondary education is a major social problem at the international level (OECD, 2022). Low success rates affect, in particular, studies in scientific disciplines, where it is especially sensitive, given the current need for professionals with a solid scientific background (European commission, 2016). Mathematics plays a particular role in this landscape: indeed, beyond the difficulties of mathematics majors, difficulties and drop-outs related to mathematics can be observed in all courses where mathematics is a service discipline. Issues related to the transition from secondary to tertiary

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education have been of interest to university mathematics education researchers for a long time and have remained a particularly active research topic, as can be seen from the contributions to the ICME14 and INDRUM 2022 conferences, the articles published in the International Journal for Undergraduate Mathematics Education (IJRUME), and the chapters in the book "Practice-Oriented Research in Tertiary Mathematics Education" (Biehler et al. 2022), to give just a few recent examples.

This ESM special issue devoted to "Transition from school into university mathematics" is therefore particularly welcome. The goal of its guest editors is to bring new results, relative to the synthesis of the literature I published in 2008 (Gueudet, 2008), in particular asking to advance research centered on affect and to expand the inclusion of other regions of the world in which the research is taking place. The 2008 synthesis was limited in both fronts, and in particular the work published at the time came almost exclusively from Europe or the USA.

In this paper, which concludes the special issue, I answer the following question: What new insights about the secondary–tertiary transition does this special issue offer?

By "new insights" I mean both new in relation to Gueudet (2008) and original. The special issue comprises seven articles. A literature review sets the scene for the other papers, providing a scientific basis for the special issue rationale (Di Martino et al. 2022b, this issue). Three papers concern students' views on the secondary–tertiary transition (Di Martino et al. 2022a; Geisler et al. 2023; Lim et al. 2023, this issue); one paper about students expressing their views as future teachers plays a transitional role (Liang et al. 2022, this issue); and the remaining two papers concern teachers' views on the secondary–tertiary transition (Kontorovich & Ovadiya, 2023; Pinto & Koichu, 2022, this issue). In the next section, I discuss new insights brought by each of the papers, followed by the contribution the papers make as a whole, highlighting the perspectives they open up in terms of future work to be carried out in the directions they indicate, or in other directions likely to complement the contributions of this special issue.

1 New insights provided by the special issue papers

When searching for new insights in a paper, different aspects can be considered, for example, its general theme and more precise focus, the research questions addressed, the theoretical frameworks, the methods used, the results stated, and the recommendations for research and for practice based on the results, to name just a few. In this special issue, all the papers concern students' or teachers' views about the secondary–tertiary transition, which, as a general theme, is new, compared with the 2008 review. With this shift, the research questions, the frameworks, and the methods are also new. So, while I consider all these elements in my analysis of the papers, and mention them briefly below, I primarily concentrate on the results obtained and on the contribution of the studies in terms of possible interventions evoked by the authors (or not evoked, but that could be designed by drawing on their results). This aligns with the perspective adopted in Gueudet (2008) where I discussed the contributions of different approaches of the secondary–tertiary transition in terms of proposed "didactical action."

1.1 Setting the scene: a literature review

The guest editors of this special issue have conducted a systematic literature review of "recent research" (Di Martino et al. 2022b, this issue), defined as research published after Gueudet (2008). Compared with the other articles in this special issue, the status of this paper is specific: it examines indeed existing research before the special issue. Hence, the question I examine here, from the perspective of the search for novelty, is: "What is new (or not) in research about secondary–tertiary transition provided by Di Martino et al. (2022b, this issue), compared with the view provided in Gueudet (2008)?" I do not attempt to make a comprehensive answer to this question but present some points that I find especially interesting. I urge the reader to read this article in detail, to discover the other aspects of this rich review.

Concerning the theoretical frameworks used in the papers, Di Martino et al. (2022b, this issue) note the presence of epistemological-cognitive approaches, used to investigate the gaps between secondary school and university mathematics and their consequences for students. This is not a new fact and is already present in the research conducted before 2008. This stability is nevertheless also linked to a change in the dominant framework they cite in the epistemological-cognitive approach, Tall's (2008) Three Worlds of Mathematics, whereas in the 2008 synthesis, the dominant framework was APOS theory (Weller et al. 2003). A similar shift is noted regarding socio-cultural theories: while the Anthropological Theory of the Didactic (Chevallard, 1992) and associated research (e.g., Bosch et al. 2004) that interprets secondary school and university as two different institutions shaping differently the mathematical content was prominent in the 2008 synthesis, and continues to be prominent, as shown in the Di Martino et al. (2022b, this issue) review, other socio-cultural frameworks with new features have emerged, in particular discursive approaches like the commognitive approach (Sfard, 2008).

Perhaps the most important evolution in terms of theoretical frameworks, in the perspective of this special issue, is the "Rite of Passage" framework (Clark & Lovric, 2009), portraying secondary–tertiary transition as entrance into a new community (of university mathematics). This rite of passage is a socio-cultural process that encompasses a cognitive shock resulting from the encounter with formal mathematics and affective dimensions. Di Martino et al. (2022b, this issue) observe that even though there are more studies using an affective perspective on secondary–tertiary transition since 2008, they are still scarce.

In the papers they reviewed, Di Martino et al. (2022b, this issue) identified four main research themes: the mathematical gap between secondary school and university, the potential of technology, factors that correlate with academic success, and the academic failure in the secondary–tertiary transition (including exam failure and dropout). The last three themes were not identified in the 2008 review and are thus new. They seem to stem from technological and social evolutions, as there has been an increase in use of digital technologies at all levels, including university, and the number of students entering scientific studies at university has doubled worldwide between 2000 and 2008 (Vieira et al. 2020), and thus student success (and failure) has become an important social problem internationally. The presence of these themes highlights the dynamism of the mathematics education research community and its concern to provide keys to understanding the current social problems.

The absence in this literature review of the theme of interventions raises questions. Has recent research only focused on understanding the existing problems, and not on proposing actions? The answer to this question should have been "no," given that in recent CERME conferences, a significant number of studies presented at the Thematic Working Group 14

about University Mathematics Education were about interventions (see e.g., González-Martín et al. 2022). These sources were not considered in Di Martino et al. (2022b, this issue), which may explain their absence. Another reason for the absence, however, could be related to the methods used by the authors, as they selected papers that describe "a transitional process, considering a before and an after of this process." This choice is very interesting and valuable, but it probably leads to missing intervention studies that are situated at either the beginning of university or at the end of secondary school, but do not really consider a before and an after. In what follows, I will continue to focus on the presence (or absence) of interventions or at least of recommendations for practice in the articles.

1.2 Students' views and experiences of secondary–tertiary transition

Three articles in this special issue investigate students' views and experiences of secondary–tertiary transition. They consider different national contexts: Germany (Geisler et al. 2023, this issue); South Korea (Lim et al. 2023, this issue); and Italy, Switzerland, and the UK (these three national contexts being compared in the paper by Di Martino et al. 2022a, this issue).

1.2.1 Explaining dropout from the first year in Germany

The study presented by Geisler et al. (2023, this issue) is motivated by what is called "the dropout problem". In Germany and internationally, a large number of students give up their scientific (including mathematics) studies during the first year. Geisler et al. (2023, this issue) belong to a body of research focusing on affective factors as a way for identifying factors that explain success or failure during the first year of university (e.g., Deshler et al. 2019; Eichler & Gradwohl, 2021; Johns, 2020). The authors use a theoretical framework combining the concepts of "Interest" and "Self-concept." They define "Interest" as a relation between a student and mathematics and "Self-concept" as the students' views on their skills and abilities. They use a survey to determine the extent to which interest in mathematics and mathematics self-concept are predictors of dropout, distinguishing between secondary school and university. Following up with a qualitative study, they identifiy situations experienced by students that have an impact on self-concept, with the aim to formulate recommendations or perhaps to design future interventions. The analyses of the students' answers to the survey show that students' initial interest in school mathematics was not related to dropout, while a low interest in university mathematics was a factor of dropout. Other factors were mathematical self-concept at the beginning of university (indeed the students who do not consider themselves as capable students in mathematics are more likely to dropout), and a negative evolution of interest, or of self-concept, during the first semester.

The authors suggest interventions likely to support a positive development of students' interest and self-concept, drawing on the results obtained in their second study. As a consequence (since the students were asked about their experience), the interventions they propose are aligned with the usual format of the German context, including additional preparatory courses and homework tasks that adapt to students' skills, making them more relevant. It is naturally always more feasible to organize interventions that can fit into the usual teaching format. Nevertheless, dealing with the critical issue of dropout may require more radical changes that cannot be designed by drawing on students' experiences in existing courses but that have to be invented by researchers.

1.2.2 Socialization into individualistic practices? The case of South Korea

Lim et al. (2023, this issue) studied students' views on secondary–tertiary transition in South Korea. They considered this transition as a socialization and enculturation process and used Cobb et al. (2001) conceptualization of norms into three categories: social norms (concerning interactions), mathematical norms (concerning the mathematics contents and the practice of mathematics), and socio-mathematical norms (concerning instruction style, and what is recognized as a valid and effective mathematical practice). They asked second and third-year university students to give examples of norms they experienced in their university and how they differed from what they experienced at secondary school. Some of their results sound quite familiar. One of the most significant differences concerns the socio-mathematical norm "Mathematical presentation begins with a theorem followed by examples including graphs and exercise problems," much more present at university. This difference aligns with the views of students in Italy, Switzerland, and the UK in Di Martino et al.'s study (2022a, this issue). Other results were more unexpected for someone more familiar with studies conducted in Western countries. According to students, the social norm "Accepting individual responsibility to develop knowledge and skills" is more present in university, compared with secondary school. Studying mathematics at university seems to be perceived by the South Korean students as a personal endeavor. Conversely, first-year students in the UK described their work with friends and their learning group as essential to meeting university expectations (Gueudet & Pepin, 2018). Another surprising result concerns the social norm "Setting the highest priority as academic excellence," which seemed to be already important at secondary school. This can be linked to the extremely high level of expectation in mathematics in secondary education in South Korea (Leung et al. 2015). So, is there really a transition problem in South Korea?

The analyses conducted in this study paint a complex picture combining academic success (the students are in second or third year) and individual suffering. The norms perceived are oriented towards individualistic practices, mathematics learning is viewed as an individualistic, competitive work, and the studying practices are driven by test scores considered as indicators of student merit. Drawing on these observations, the authors recommend the design and implementation of interventions at university. They propose that these interventions are oriented towards a better balance between individual knowledge acquisition and collective participation. In my opinion, these recommendations are not directly linked to the secondary–tertiary transition but to the kind of mathematical practices that we want our students to develop. Academic success is not the only criterion for a successful transition; it must also lead to the mathematical practices that we wish to develop in students.

1.2.3 An international comparison in Europe

Di Martino et al. (2022a, this issue) presented a cross-cultural study of students' experience of secondary–tertiary transition taking place in three European countries: Italy, Switzerland, and the UK. Using the rite of passage framework (Clark & Lovric, 2009) and the three-dimensional model for attitude toward mathematics (Di Martino & Zan, 2010), they investigated students' change in attitude during the secondary–tertiary transition. They found a difficult situation. As university level students' perceived competence declined, negative emotional dispositions towards mathematics appeared. These evolutions were

linked to negative experiences, such as repeated exam failures or stress induced by the fast pace of lectures. However, the comparative nature of the study uncovered differences between the situations in the different countries: e.g., the new requirements for proof seem to be of concern especially for the Swiss students; the presence of other high-achieving students seems especially concerning for Italian students.

Investigating secondary–tertiary transition with a cross-cultural approach and perspective is an original and promising direction for research. Only a few studies (of which I am aware) attempted this so far. For example, Alves Dias et al. (2010) conducted a curricular comparison at the secondary–tertiary transition concerning functions in Brazil and in France, as a preliminary step for developing digital resources for supporting this transition. They documented major differences between the curricula, likely to lead to different mathematical practices at the beginning of university for Brazilian and French students. In a similar vein, Di Martino et al. (2022a, this issue) observed that the secondary school curricula evoked by students differ from one country to another, impacting their experience at university. These results suggest that a preliminary comparative analysis of the curricula in the three countries can usefully inform a further study. The authors identified other national particularities linked to other aspects of educational policies in the three countries, such as the change in the class size between secondary school and university, or a new selection process creating classes of high-achieving students.

Di Martino et al. (2022a, this issue) confirmed that the secondary–tertiary transition is indeed a crisis for students in different countries and that there are major national differences beyond this international problem. Is it possible to build on these results to design interventions? It seems that the differences are linked to national educational policies, such as the secondary school curricula, how students enter university, and how orientation choices for the first year at university are linked to a professional project. Identifying some features of a secondary school mathematics curriculum likely to promote a successful transition to university physics courses, for example, can be a promising direction for future research that could perhaps inform stakeholders and educational authorities in different countries.

1.3 Double discontinuity and students' experiences

Research interest in the training at university of future mathematics teachers has developed widely since 2008, in particular with a large number of studies referring to Klein's double discontinuity (Klein, 1908/1939) (see e.g., Barquero & Winsløw, 2022; Kilpatrick, 2019; Viirman, 2022). While such interest can be seen as a new direction for research about secondary–tertiary transition (which is the first transition in Klein's model), most studies referring to double discontinuity focus primarily on the second transition and investigate how mathematical knowledge learned at university can nurture the teaching practice of novice secondary school teachers.

The paper by Liang et al. (2022, this issue) makes an original contribution in terms of Klein's (1908/1939) double discontinuity, bringing some new ideas about the first transition. They investigate prospective teachers' views on secondary–tertiary transition in terms of their own recent experience, as they are in their third year of university in Hong Kong, and on what they plan to teach to their future secondary school students to prepare them for their mathematics university studies, that is, their intentions to address the secondary–tertiary transition as future teachers. Liang et al. adapted the concept of figured world (Holland et al. 1998) to the context of mathematics education research. In their

study, the double discontinuity is conceptualized as a change of figured worlds related to secondary mathematics, university mathematics, and teaching mathematics at secondary school. The authors complement this framework with the three-dimensional model of attitudes (TMA, Di Martino & Zan, 2010). Their analysis of their interviews provides interesting results about these students' own secondary–tertiary transition experience. The most original aspect of the study, in my opinion, comes from the analysis of lesson plans that the students were asked to design for potential secondary school students with the aim to address as teachers the difficulties, or at least the changes that they met in secondary–tertiary transition. Overall, in these lessons, it seemed that their goal was to approach more complex and elaborate mathematics than they themselves had encountered in high school. This introduces an original view of the double discontinuity, as a challenge with possible positive outcomes rather than a deficit. Will these students, once they become secondary school teachers, really implement such practices? This is very unlikely, due to the institutional constraints framing teaching practices. Nevertheless, this paper opens an interesting perspective, in terms of future research and possible interventions. Based on my experience as a teacher educator, it is apparent that the question of how secondary school teachers can adequately prepare their students for university mathematics is not yet a central theme in preservice teacher education. However, understanding how such preparation could be included in teacher education courses deserves future investigation.

1.4 Teachers' views on transition

Two articles in this special issue concern the views of university teachers on secondary–tertiary transition. Kontorovich and Ovadiya (2023, this issue) address the views of tutors about students' difficulties and possible pedagogical strategies whereas Pinto and Koichu (2022, this issue) analyze the views of university teachers concerning students' difficulties and focuses on possible ways to improve the secondary–tertiary transition.

Kontorovich and Ovadiya investigate how the knowledge and practices of tutors in a New Zealand university evolve over time, using their pedagogical discourse as unit of analysis. The authors refer to the commognitive approach (Sfard, 2008), one of the socio-cultural theoretical frameworks increasingly used for the study of university mathematics education and in particular of secondary–tertiary transition (Nardi, 2019).

Using texts the tutors produced as part of their mathematics education course, the authors identify incidents that describe first-year students' unexpected difficulties. They note that the narratives are deficit oriented (see Pinto & Koichu, 2022, this issue) like those of experienced university teacher. I wonder if the interpretation here could be more psychological than socio-cultural, as secondary school teachers propose the same kind of argument to explain the difficulties of their students in the primary-secondary transition (see e.g., Prendergast et al. 2019). More than a narrative from a specific community, it may be a human characteristic to attribute the cause of the failure of a given task to those who undertook that task before them.

Beyond this initial deficit discourse, Kontorovich and Ovadiya's (2023, this issue) study evidences positive evolutions: the tutors progressively deepen their interpretations of students' difficulties, and develop strategies to face these difficulties. So, this paper presents a success story: that of a mathematics education course which contributed to the tutors' (early) professional development and to their sense-making of secondary–tertiary transition. It aligns with the results obtained by Grove and Croft (2019) who observed the learning of tutors in a Mathematics Support Center and identified their discussions with other

tutors as a central factor in such learning. These results suggest possible interventions oriented towards the creation of collectives enabling tutors—or ideally all university teachers—to deepen their pedagogical reflection by sharing their practices with their peers or with teacher educators.

Using responses collected in the 2019 international survey[1] of university teachers on the secondary–tertiary transition and their perceptions of the change in the last ten years, Pinto and Koichu (2022, this issue) investigated university teachers' views of ways to improve the transition. The responses were provided by 310 instructors from 30 different countries.

The authors classified the propositions (including both reforms and projects) into four categories: *jump-oriented*, when the focus is on the gap between secondary and tertiary education; *enculturation-oriented* when secondary school and university are viewed as two different institutions or communities with different cultures; *diversity-oriented* when the focus is on the diversity of the students; and *cross-community oriented* for suggestions of collaboration between different groups of teachers (typically secondary school and university teachers). Pinto and Koichu's (2022, this issue) study offers many interesting and thought-provoking results. Some of these findings are on the alarming side: most respondents say that the difficulties linked with secondary–tertiary transition have been aggravated in the past 10 years. They consider that these difficulties are the result of inadequate student preparation and recommend reforms concerning secondary school. Other findings offer a more positive view: some teachers acknowledge the diversity of the students. They value diversity-oriented measures, enculturation-oriented measures like tutoring, and call for more professional development at university level.

While the question of possible interventions improving the secondary–tertiary transition is not new, collecting the views of an international sample of university teachers is new. In line with the cross-cultural view promoted in this special issue (see e.g., Di Martino et al. 2022a, b, this issue), it would be interesting to know if the analysis of the answers reveals national or regional specificities concerning the kind of initiatives already implemented or considered desirable. In Pinto and Koichu's (2022, this issue) study, the focus on an international sample seems to be aimed at collecting a variety of responses but not at identifying such specificities. Are the different types of initiatives equally distributed, or are some types cited only by teachers from certain countries? Knowing the answer to this question is important for studying the factors that lead to certain kinds of responses, especially if this research is to inform future interventions.

I also observe that the two kinds of initiatives, *diversity-oriented* and *cross-community oriented*, were not evoked in Gueudet (2008). In addition, this synthesis presented interventions proposed by research studies, and we do not know if the university faculty at the time were aware of these suggested interventions. The results of Pinto and Koichu's (2022, this issue) study seem to indicate a progress in faculty's pedagogical awareness, or even in the communication between mathematics educators and mathematicians. Nevertheless, the respondents to the survey did not represent the whole university teaching community: they were interested enough in their teaching activity to dedicate time for answering an online survey. Collecting the views of a representative sample of university teachers is a real challenge for research, especially for research about secondary–tertiary transition. In fact, a representative sample should include colleagues who do not want to teach first-year students but are compelled to do so by their institution. According to my personal experience

[1] This survey has been organized by the educational committee of the European Mathematical Society.

in France, a sizable percentage of my colleagues fall within this category. However, the observation of a real pedagogical reflection and the recognition of a need for training among the colleagues who answered the survey in Pinto and Koichu's (2022, this issue) study open interesting avenues for interventions, and a challenge for mathematics education researchers.

2 Discussion and perspectives for future research

So finally, after reading these seven papers, what did we learn that was new? And what was missing, or more positively what perspectives are opened by this special issue for further research? I noted the new theoretical frames used; they can be socio-cultural and affect-oriented, and in this special issue, the intention was to combine these two perspectives. I also noted new results; some of them are linked to the expansion of authors' regional origins and to highlighting differences between different countries. More generally in my view, the most important contribution of this special issue are results evidencing the complexity and diversity of the secondary–tertiary transition. Finally, I observed new directions concerning interventions suggested by the studies (explicitly mentioned or not). I deepen and complement below the reflection on these different directions.

2.1 Affects or/and socio-cultural issues? Theoretical perspectives

In Gueudet (2008), I structured the presentation of the studies according to the theoretical frameworks adopted, which framed authors' perspective on secondary–tertiary transition. This special issue, despite its more precise focus on affect-oriented and socio-cultural approaches of secondary–tertiary transition, shows that this entry by theoretical frameworks and how they frame the studies on secondary–tertiary transition is relevant. Indeed, a significant diversity of theoretical frameworks is present in the articles. Four of them concern affect:

- The "Rite of passage" framework is used in Di Martino et al. (2022a, this issue).
- The "Three-dimensional model of attitudes" framework is used in two papers: Di Martino et al. (2022a, this issue) and Liang et al. (2022, this issue).
- The "Figured world" framework is used by Liang et al. (2022, this issue).
- The "Interest" and "Self-concept" frameworks are used by Geisler et al. (2023, this issue).

Other authors use frameworks pertaining to socio-cultural approaches not directly linked with affect, such as the following:

- The commognitive theory (Kontorovich & Ovadiya, 2023, this issue)
- The framework of social norms and enculturation (Lim et al. 2023, this issue)

This description may suggest that some papers focus only on affect, while others consider socio-cultural issues. However, the situation is more complex because the "Rite of Passage" concerns affect, but it is itself a socio-cultural framework. The link between affect (an individual student experience) and socio-cultural perspectives (which look at the influence of the social and/or institutional environment) suggests that the collectively shared

experience of secondary–tertiary transition is a complex theoretical issue. This theoretical tension may lead to different recommendations in terms of interventions for supporting secondary–tertiary transition, ranging from individual tutoring that may reduce stress to collective student work that may help in the understanding and integration of social norms.

Some papers in this special issue used two different frameworks, but the authors did not explicitly draw on theoretical networking strategies (Bikner-Ahsbahs & Prediger, 2010). Investigating more systematically how socio-cultural approaches and approaches focusing on students' (potentially individual) affect can be networked could and should be developed in further research. For example, is it possible to coordinate the framework of the didactic contract (socio-cultural perspective used in several studies about secondary–tertiary transition, see e.g., Pepin, 2014) with a perspective that accounts for students' and teachers' beliefs? Similarly, can the communities of practice framework (some studies consider secondary–tertiary transition as entry in a new community of practice, see e.g., Biza et al. 2014) be coordinated with a perspective that accounts for students' self-efficacy? These complex theoretical questions are associated with methodological issues. The studies about secondary–tertiary transition using an affect-oriented framework often rely on methods based on students' or teachers' answers to surveys or in the context of interviews. A study using a community of practice framework (a socio-cultural theoretical framework) requires the identification of the practice and direct observations of such practice; in the case of secondary–tertiary transition, it would require observations and/or videos of lectures or tutorials. These kinds of data are not present in this special issue's papers. Thus, I note here two associated needs for future research about secondary–tertiary transition: explicit theoretical networking between socio-cultural and affect-oriented theories and the collection and triangulation of participants' statements with their actual practice.

2.2 International view and research about secondary–tertiary transition

One of the statements motivating this special issue, confirmed by the literature review (Di Martino et al. 2022b, this issue), is that published research on secondary–tertiary transition mostly comes from Europe and from the USA. This statement leads the guest editors to note a limitation concerning the international view of secondary–tertiary transition. Indeed, considering secondary–tertiary transition with a socio-cultural perspective raises our awareness about local, national, or regional specificities, and the papers in this special issue evidence such specific characteristics. The special issue includes papers from South Korea, Hong-Kong, and New Zealand in addition to papers from Europe. The articles do show important national specificities, which impact the secondary–tertiary transition and the recommendations to overcome difficulties. For example, in South Korea, students work hard at secondary school to prepare for the national college entrance exam and they are selected in colleges according to their score in mathematics, whereas in Germany, some students enter scientific courses at university with deficiencies regarding their mastery of secondary school content. This can explain the popularity of bridge courses among German universities, but such courses would not be offered in South Korea. Studies such as Di Martino et al. (2022a, this issue) make the point about the importance of truly understanding the curriculum that students are exposed to in secondary schools because they can explain substantial cross-cultural differences and practices. While such research is challenging, as they require translations and negotiation of meanings across researchers and participants, these studies offer richer perspectives on the phenomenon of secondary–tertiary transition.

The statement, "published research on secondary–tertiary transition comes from only some parts of the world" (which unfortunately also holds for published research in mathematics education at large), can also lead to another conclusion: What is missing is an international view of *research about* secondary–tertiary transition? What are the research foci of interest in the different parts of the world? It is not clear to me whether the guest editors intended to contribute to answering this question. The special issue brings some points of reflection. For example, Lim et al.'s (2023, this issue) paper focuses on the socialization experience of students into individualistic and competitive practices in South Korea, but I am not aware of other published papers with a similar focus. Another interesting point of reflection comes from the notable absence in this special issue of papers coming from the USA. In undergraduate mathematics education journals and books, articles and chapters by US authors are often dominant. Indeed 56% of the 64 papers published in IJRUME between 2019 and 2021 were from the USA, suggesting that the research foci from the USA research community are widely present in the literature. Thus, their absence in this special issue makes particularly the impact of missing one part of the world on our vision of research visible. For example, equity-oriented teaching of university mathematics is increasingly researched in the USA (e.g., Adiredja & Andrews-Larson, 2017), yet such focus on equity is not present in this special issue. Rather than criticism, this point should be an opportunity to highlight in a very concrete way the consequences of not having work that comes from one region of the world. Naturally, it is especially visible when the missing region includes North America, because of the intense research activity from the RUME[2] community on these topics. We need to continue the collective effort[3] to develop the access to communication and publication for authors from all parts of the world; this is a major challenge for the mathematics education research community.

2.3 Complexity and diversity of secondary–tertiary transition experience

Student access to scientific studies at university has increased dramatically since the years 2000s (Vieira et al. 2020) and their failure or success has become an international social concern. Di Martino et al. (2022b, this issue) evidenced that the factors explaining success or failure are now important themes of research about secondary–tertiary transition. As mentioned earlier, this evolution of research evidences how the international community of research in mathematics education is able to take hold of social problems and to search for understanding and solutions. This access to university studies for a larger number of students has also led to a greater diversity of these students. Thus, research has also evolved in focus to understand the consequences of this greater diversity.

This special issue shows how research has taken this diversity into account and documents progress made in this direction. The literature published before 2008 often considered a generic student, and Gueudet (2008) conveyed the idea of a single international transition problem. The articles in this special issue have advanced greatly from this simplistic view. The teaching and learning phenomena are complex and acknowledging this complexity is an important contribution of any educational research (Artigue, 2021). Pinto and Koichu (2022, this issue) identify the recent emergence in the research literature of studies evidencing a form of "diversification of secondary–tertiary transition," recognizing that

[2] Research in Undergraduate Mathematics Education, http://sigmaa.maa.org/rume/Site/News.html

[3] See in particular the "Joint statement to reduce reviewer bias" https://mathematicseducationjournals.com/

secondary–tertiary transition is not the same for all students (even in the same university) and that it may be experienced differently by different students. The articles in this special issue confirm this diversity. Collectively, they offer a rich view of what the experience of some students can be, especially when they draw on interviews with individual students (e.g., Geisler et al. 2023; Kontorovich & Ovadiya, 2023; Liang et al. 2022, in this issue).

This does not mean that the secondary–tertiary transition experience should be seen as an individual phenomenon and studied as such; the articles also show the importance of the social dimension of the phenomenon. The individual experiences do not just replace the view of secondary–tertiary transition as a single problem with a view as an inextricable set of diverse individual situations. Instead, they highlight the value of investigating the different experiences of different groups of students. This is a promising direction for future research and also a requirement for designing, implementing, and evaluating interventions.

2.4 Towards the design of affect-oriented interventions?

In my view, the papers in this special issue suggest different kinds of interventions for supporting a successful secondary–tertiary transition; I summarize them here:

- Development of support initiatives like bridge courses, reinforcing the students' preparation but also their self-concept (Geisler et al. 2023, this issue)
- Development of courses for future secondary school teachers leading them to reflect on a better preparation for their future students (Liang et al. 2022, this issue)
- Development of professional development for tutors (Kontorovich & Ovadiya, 2023, this issue)
- Development of professional development for university teachers, in particular by creating groups of secondary school and university teachers who work and discuss together and improve this way their mutual understanding (Pinto & Koichu, 2022, this issue)

Lim et al. (this issue) call for more work of students in small groups, with the aim of developing cooperation in mathematics courses at university. The studies by Liang et al. (2022, this issue) and Kontorovich and Ovadiya (this issue) implemented courses, for future secondary school teachers and for tutors, respectively. Nevertheless, their studies do not investigate the design of these courses, nor do they evaluate their impact in terms of improvement of students' experience during the secondary–tertiary transition.

Schukajlow et al.'s (2017) survey of the literature about emotions and motivation in mathematics education research notes that intervention studies targeting a change in student affect (e.g., increasing motivation and engagement) exist, but are scarce. This observation seems to apply for this special issue, too; indeed, none of the studies investigates the systematic design, implementation, and evaluation of courses that could support the development of first-year students' self-efficacy, for example. Recent research about inquiry-based mathematics education or other forms of students' active learning (see e.g., Laursen & Rasmussen, 2019) suggests that this kind of teaching can positively influence students' mathematical attitudes and identities (e.g., Kober, 2015; Voigt et al. 2022). While those authors do not directly focus on secondary–tertiary transition, their findings support the hypothesis that interventions targeting a change in students' affect during the first year at university are a promising direction for future research.

The secondary–tertiary transition problem has worsened in recent years. Our colleagues teaching at tertiary level, mathematicians, and colleagues from other scientific disciplines

such as physics, economics, and computer science have become aware of this problem. Producing research results about secondary–tertiary transition is a challenge, but also an opportunity in terms of relations with these colleagues. This special issue provides a better understanding of the complex and diverse transition problem, and contributes to meeting this challenge.

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Declarations

Competing interests The author declares no competing interests.

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